



Problems (Set Theory)

1. Which of the following are sets? Justify your answer.
 - a. The collection of all the months of a year beginning with the letter J.
 - b. The collection of ten most talented writers of India.
 - c. A team of eleven best-cricket batsmen of the world.
 - d. The collection of all boys in your class.
 - e. The collection of all natural numbers less than 100.
 - f. A collection of novels written by the writer Munshi Prem Chand.
 - g. The collection of all even integers.
 - h. The collection of questions in this Chapter
 - i. A collection of most dangerous animals of the world
2. Write the following sets in roster form:
 - a. $A = \{x : x \text{ is an integer and } -3 \leq x < 7\}$
 - b. $B = \{x : x \text{ is a natural number less than 6}\}$
 - c. $C = \{x : x \text{ is a two-digit natural number such that the sum of its digits is 8}\}$
 - d. $D = \{x : x \text{ is a prime number which is divisor of 60}\}$
 - e. $E = \text{The set of all letters in the word TRIGONOMETRY}$
 - f. $F = \text{The set of all letters in the word BETTER}$
3. Write the following sets in the set-builder form:
 - a. $\{3, 6, 9, 12\}$
 - b. $\{2, 4, 8, 16, 32\}$
 - c. $\{5, 25, 125, 625\}$
 - d. $\{2, 4, 6, \dots\}$
 - e. $\{1, 4, 9, \dots, 100\}$
4. Which of the following are examples of the null set
 - a. Set of odd natural numbers divisible by 2
 - b. Set of even prime numbers
 - c. $\{x : x \text{ is a natural numbers, } x < 5 \text{ and } x > 7\}$
 - d. $\{y : y \text{ is a point common to any two parallel lines}\}$
5. Examine whether the following statements are true or false:
 - a. $\{a, b\} \not\subset \{b, c, a\}$
 - b. $\{a, e\} \subset \{x : x \text{ is a vowel in the English alphabet}\}$
 - c. $\{1, 2, 3\} \subset \{1, 3, 5\}$
 - d. $\{a\} \subset \{a, b, c\} \vee \{a\} \in \{a, b, c\}$

- e. $\{x : x \text{ is an even natural number less than } 6\} \subset \{x : x \text{ is a natural number which divides } 36\}$
6. Let $A = \{1, 2, \{3, 4\}, 5\}$. Which of the following statements are incorrect and why?
- $\{3, 4\} \subset A$
 - $\{3, 4\} \in A$
 - $\{\{3, 4\}\} \subset A$
 - $1 \in A$
 - $1 \subset A$
 - $\{1, 2, 5\} \subset A$
 - $\{1, 2, 5\} \in A$
 - $\{1, 2, 3\} \subset A$
 - $\varphi \in A$
 - $\varphi \subset A$
 - $\{\varphi\} \subset A$
7. Write down all the subsets of the following sets (i) $\{a\}$ (ii) $\{a, b\}$ (iii) $\{1, 2, 3\}$ (iv) φ
8. Write the following as intervals: (i) $\{x : x \in \mathbb{R}, -4 < x \leq 6\}$ (ii) $\{x : x \in \mathbb{R}, -12 < x < -10\}$ (iii) $\{x : x \in \mathbb{R}, 0 \leq x < 7\}$ (iv) $\{x : x \in \mathbb{R}, 3 \leq x \leq 4\}$
9. Write the following intervals in set-builder form: (i) $(-3, 0)$ (ii) $[6, 12]$ (iii) $(6, 12]$ (iv) $[-23, 5)$
10. If $A = \{x : x \text{ is a natural number}\}$, $B = \{x : x \text{ is an even natural number}\}$, $C = \{x : x \text{ is an odd natural number}\}$ and $D = \{x : x \text{ is a prime number}\}$, find (i) $A \cap B$ (ii) $A \cap C$ (iii) $A \cap D$ (iv) $B \cap C$ (v) $B \cap D$ (vi) $C \cap D$
11. If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{2, 4, 6, 8\}$ and $B = \{2, 3, 5, 7\}$. Verify that (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$
12. Decide, among the following sets, which sets are subsets of one and another: $A = \{x : x \in \mathbb{R} \text{ and } x \text{ satisfy } x^2 - 8x + 12 = 0\}$, $B = \{2, 4, 6\}$, $C = \{2, 4, 6, 8, \dots\}$, $D = \{6\}$.
13. In each of the following, determine whether the statement is true or false. If it is true, prove it. If it is false, give an example.
- If $x \in A$ and $A \in B$, then $x \in B$
 - If $A \subset B$ and $B \in C$, then $A \in C$
 - If $A \subset B$ and $B \subset C$, then $A \subset C$
 - If $A \not\subset B$ and $B \not\subset C$, then $A \not\subset C$
 - If $x \in A$ and $A \not\subset B$, then $x \in B$
 - If $A \subset B$ and $x \notin B$, then $x \notin A$
14. Let A , B , and C be the sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$. Show that $B = C$.
15. Show that the following four conditions are equivalent: (i) $A \subset B$ (ii) $A - B = \varphi$ (iii) $A \cup B = B$ (iv) $A \cap B = A$
16. Show that if $A \subset B$, then $C - B \subset C - A$.
17. Show that for any sets A and B , $A = (A \cap B) \cup (A - B)$ and $A \cup (B - A) = (A \cup B)$
18. Using properties of sets, show that (i) $A \cup (A \cap B) = A$ (ii) $A \cap (A \cup B) = A$.

19. Show that $A \cap B = A \cap C$ need not imply $B = C$.
20. Let A and B be sets. If $A \cap X = B \cap X = \varnothing$ and $A \cup X = B \cup X$ for some set X , show that $A = B$.
21. Find sets A , B and C such that $A \cap B$, $B \cap C$ and $A \cap C$ are non-empty sets and $A \cap B \cap C = \varnothing$.
22. Find the following cardinalities
 - a. $|\{x \in \mathbb{Z} : |x| < 10\}|$
 - b. $|\{x \in \mathbb{N} : x^2 < 10\}|$
23. Sketch the following in XY plane
 - a. $\{(x, y) : x \in [1, 2], y \in [1, 2]\}$
 - b. $\{(x, y) : |x| = 2, y \in [0, 1]\}$
 - c. $\{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 = 1\}$
 - d. $\{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 \leq 1\}$
 - e. $\{(x, x+y) : x \in \mathbb{R}, y \in \mathbb{Z}\}$
24. Suppose $A = \{0, 1\}$ and $B = \{1, 2\}$, Find
 - a. $P(A) - P(B)$
 - b. $P(A) \cap P(B)$